IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Sumant Chakravarty
25.02.2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology

- Collected launch data from the SpaceX REST API and scraped Falcon 9 launch tables from Wikipedia
- Cleaned the data, imputed missing payload mass and created the binary landing label (Class)
- Explored the data with SQL, Seaborn/Matplotlib, Folium maps and a Plotly Dash dashboard
- Trained Logistic Regression, SVM, Decision Tree and KNN, tuned with GridSearchCV (10-fold CV) and evaluated with accuracy and confusion matrices

Results

- **Success rate:** 66.7% of 90 Falcon 9 first stages landed
- **Experience:** 0% landed in 2010–2013, ~83% in 2017, a dip to ~61% in 2018, then ~90% in 2019
- **Orbits:** VLEO (86%) and LEO (71%) do best; GTO is lowest (52%) among orbits with enough launches
- **Sites:** KSC LC-39A has the highest success rate; CCAFS SLC-40 hosts most launches
- **Models:** all four reach 83.3% test accuracy (15/18); Decision Tree has the best CV accuracy (88.8%)

Introduction

SpaceX advertises Falcon 9 launches at $62 million, compared with $165 million or more from other providers. Much of the saving comes from reusing the first stage, but not every booster is recovered: some crash, and some are sacrificed because of payload, orbit or customer requirements.

As a data scientist at Space Y, a fictional competitor founded by billionaire Allon Musk, my task is to:

- Predict whether the Falcon 9 first stage will land successfully
- Use that prediction to estimate the cost of a launch
- Build dashboards and machine learning models that support bidding decisions

Key question: Which launch characteristics determine whether the first stage lands successfully?

Section 1

Methodology

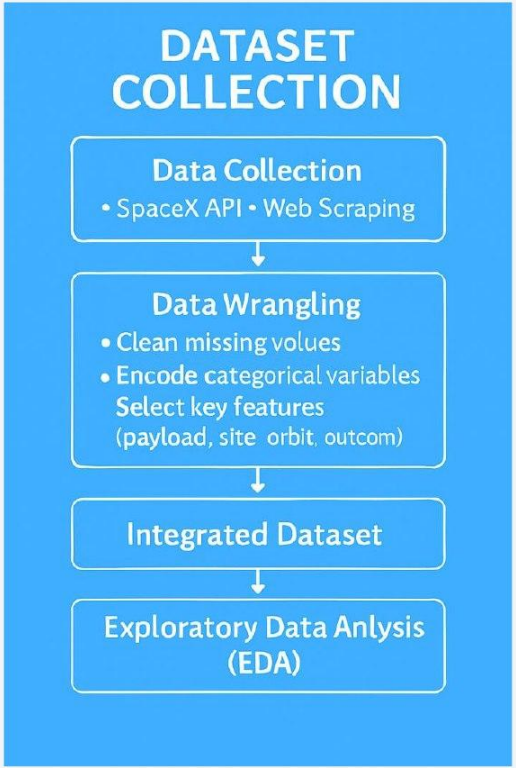


Methodology

- **Data collection:** SpaceX REST API, supplemented by web scraping Wikipedia launch tables
- **Data wrangling:** filtered to Falcon 9, imputed missing payload mass with the mean, converted landing outcomes into a binary label
- **Exploratory analysis:** SQL queries on SQLite plus Seaborn/Matplotlib charts
- **Interactive analytics:** Folium maps of launch sites and their surroundings; Plotly Dash dashboard
- **Predictive analysis:** Logistic Regression, SVM, Decision Tree and KNN; hyperparameters tuned with GridSearchCV (10-fold CV); evaluated with test accuracy and confusion matrices

Data Collection

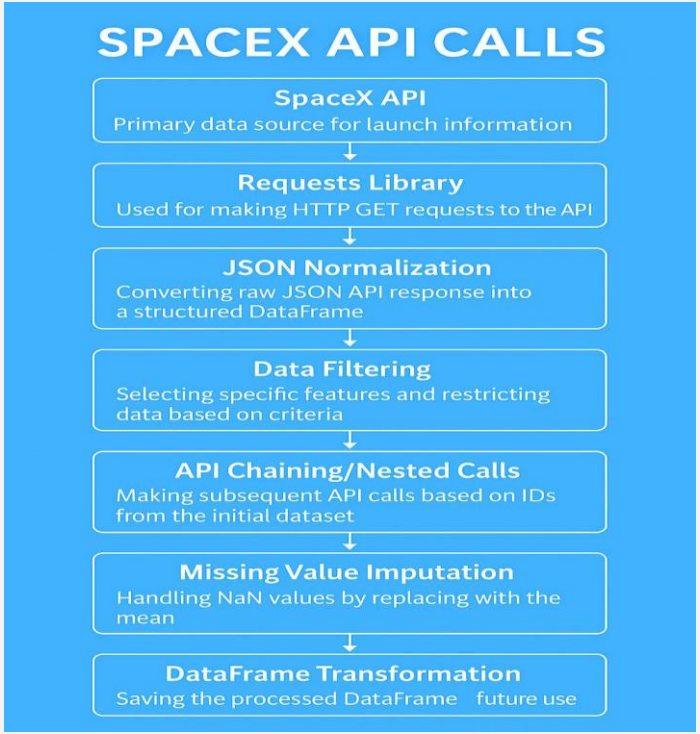
- Retrieved launch data from the **SpaceX API**.
- Supplemented with additional information via **web scraping**.
- Consolidated datasets using **Pandas** for structured analysis.



Data Collection – SpaceX API

- Requested past launches from the SpaceX REST API (api.spacexdata.com/v4/launches/past)
- Resolved rocket, launchpad, payload and core IDs with follow-up API calls
- Normalized the JSON into a DataFrame and kept Falcon 9 launches only
- Replaced missing payload mass values with the mean

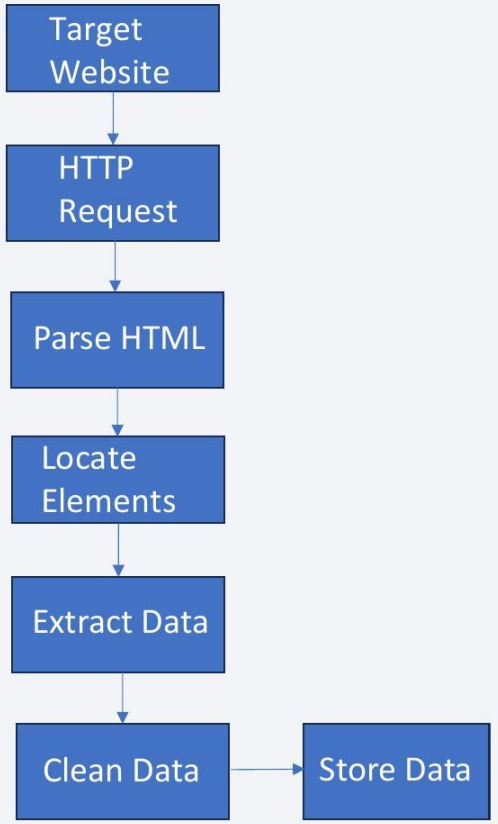
- **Notebook:** notebooks/01_data_collection_api.ipynb



Data Collection - Scraping

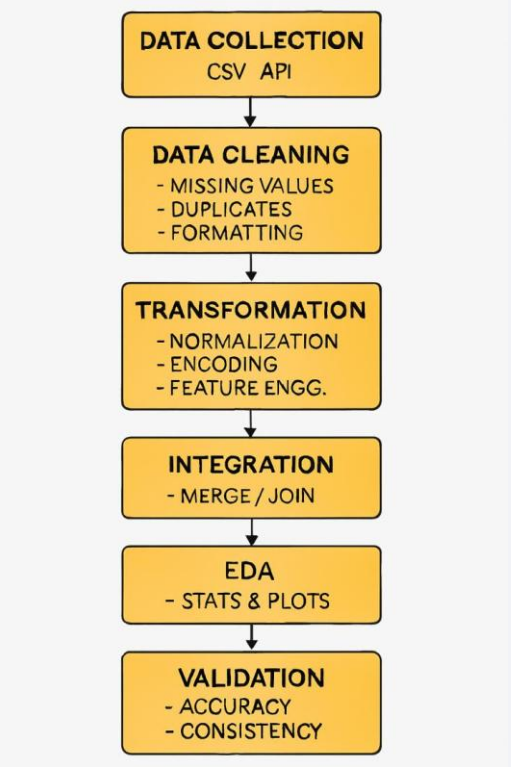
- Requested a fixed snapshot of the Wikipedia page "List of Falcon 9 and Falcon Heavy launches"
- Parsed the HTML with BeautifulSoup and located the launch tables
- Extracted column names from the table headers
- Parsed each row into flight number, date, booster, site, payload, orbit, customer and outcomes
- Built a DataFrame and saved it to CSV

- **Notebook:** notebooks/02_data_collection_webscraping.ipynb



Data Wrangling

- Calculated the share of missing values per column (LandingPad: 28.9% missing, by design for launches without a landing pad)
- Counted launches per site and per orbit
- Grouped the landing outcomes and marked unsuccessful ones (False ASDS, False Ocean, False RTLS, None ASDS, None None)
- Created the Class label: 1 = landed, 0 = did not land
- Overall success rate: 66.7%
- **Notebook:** notebooks/03_data_wrangling.ipynb



EDA with Data Visualization

- **Scatter plots** of Flight Number and Payload Mass against Launch Site and Orbit, coloured by outcome, to see how experience, payload and destination relate to landing success
- **Bar chart** of success rate by orbit type, to compare orbits directly
- **Line chart** of average success rate by year, to show the trend over time
- **Feature engineering:** one-hot encoded Orbit, LaunchSite, LandingPad and Serial, and cast all features to float64 for modelling
- **Notebook:** notebooks/05_eda_visualization.ipynb

EDA with SQL

Queries run on the SPACEXTABLE (SQLite)

- Distinct launch sites
- First 5 records from sites starting with 'CCA'
- Total payload mass for NASA (CRS)
- Average payload mass for booster F9 v1.1
- Date of the first successful ground-pad landing
- Boosters with drone-ship success and payload of 4,000–6,000 kg
- Count of successful and failed mission outcomes
- Boosters that carried the maximum payload (subquery)
- Drone-ship landing failures in 2015, by month
- Landing outcomes ranked by count, 2010-06-04 to 2017-03-20

- **Notebook:** notebooks/04_eda_sql.ipynb

Build an Interactive Map with Folium

- **Circles and markers** label each launch site and NASA Johnson Space Center
- **Marker clusters** show every launch at a site, coloured green (landed) or red (did not land)
- **Mouse position** plugin to read coordinates of nearby coastline, railway, highway and city
- **Polylines** draw the distance from a launch site to each of those points
- **Notebook:** notebooks/06_launch_site_analysis_folium.ipynb

Build a Dashboard with Plotly Dash

- **Launch-site dropdown** to view all sites or a single site
- **Pie chart:** share of successful launches by site (all sites), or success vs. failure for the selected site
- **Payload range slider** to filter launches by payload mass
- **Scatter chart:** payload mass vs. landing outcome, coloured by booster version, to show how payload and booster type relate to success
- **App:** dashboard/spacex_dash_app.py

Predictive Analysis (Classification)

1 Prepare

Class column as target (Y); standardized features (X); 80/20 train/test split (18 test launches)

2 Tune

GridSearchCV with 10-fold CV for Logistic Regression, SVM, Decision Tree and KNN

3 Evaluate

CV accuracy (best_score_), test accuracy (.score) and a confusion matrix per model

4 Select

All models tie on test accuracy, so CV accuracy breaks the tie: Decision Tree (0.888)

- **Notebook:** notebooks/08_machine_learning_prediction.ipynb

Results

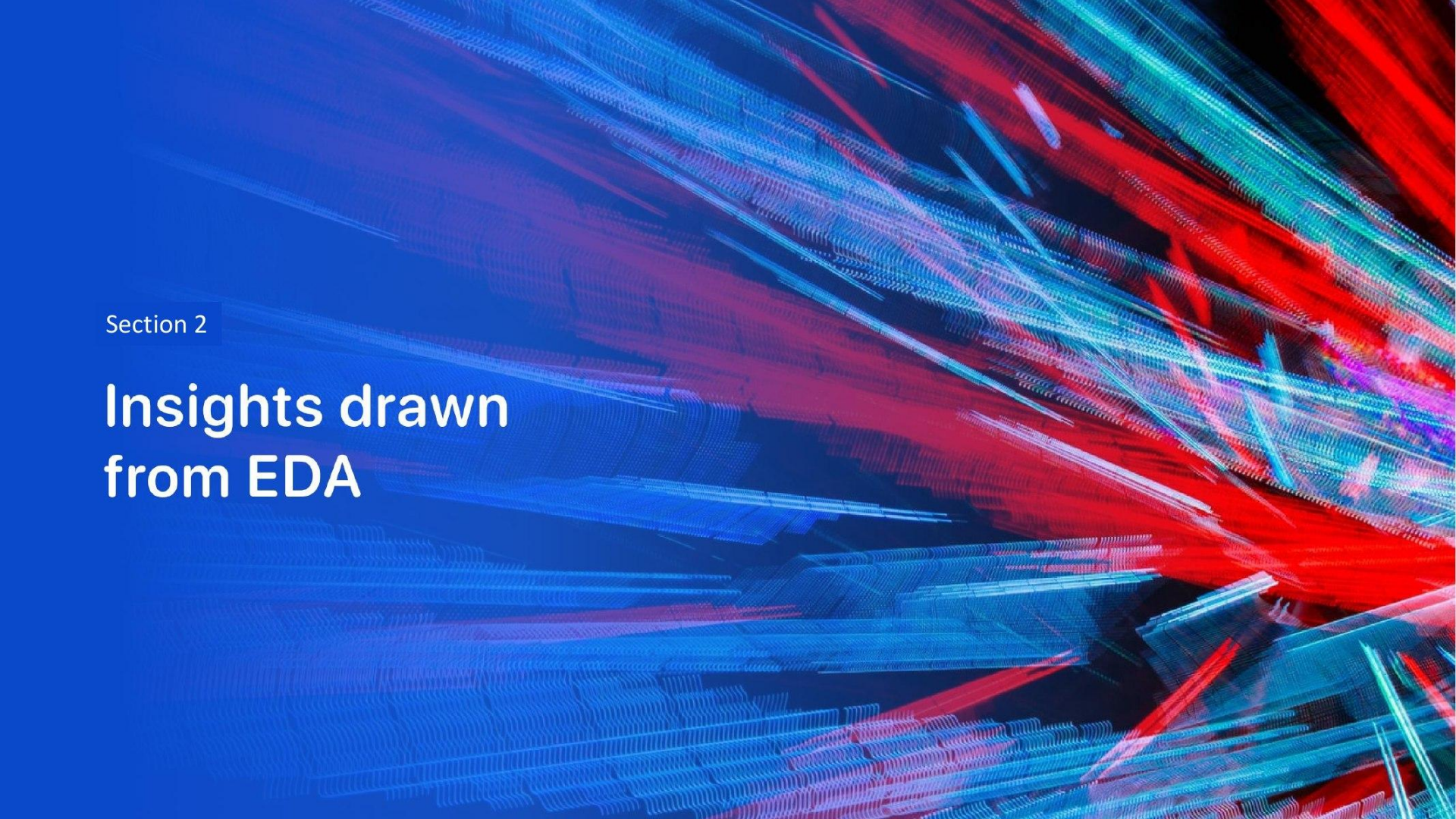
Model performance

Model	CV accuracy	Test accuracy
Logistic Regression	0.846	0.833
SVM	0.848	0.833
Decision Tree	**0.888**	0.833
KNN	0.848	0.833

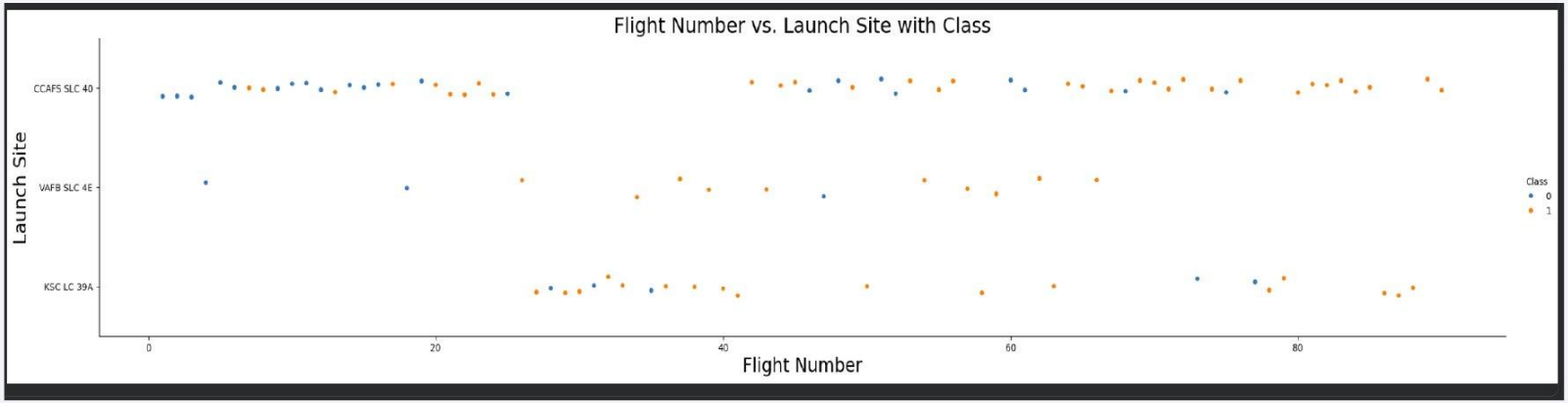
All models score the same on the 18-launch test set; the Decision Tree has the best cross-validation accuracy.

Exploratory analysis

- **90 launches;** 66.7% of first stages landed
- **Launches per site:** CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13
- **Most common orbits:** GTO 27, ISS 21, VLEO 14, PO 9
- **Outcomes:** True ASDS 41, None None 19, True RTLS 14, False ASDS 6, True Ocean 5, False Ocean 2, None ASDS 2, False RTLS 1
- **Missing data:** only LandingPad (28.9%), by design


Section 2
Insights drawn
from EDA

Flight Number vs. Launch Site



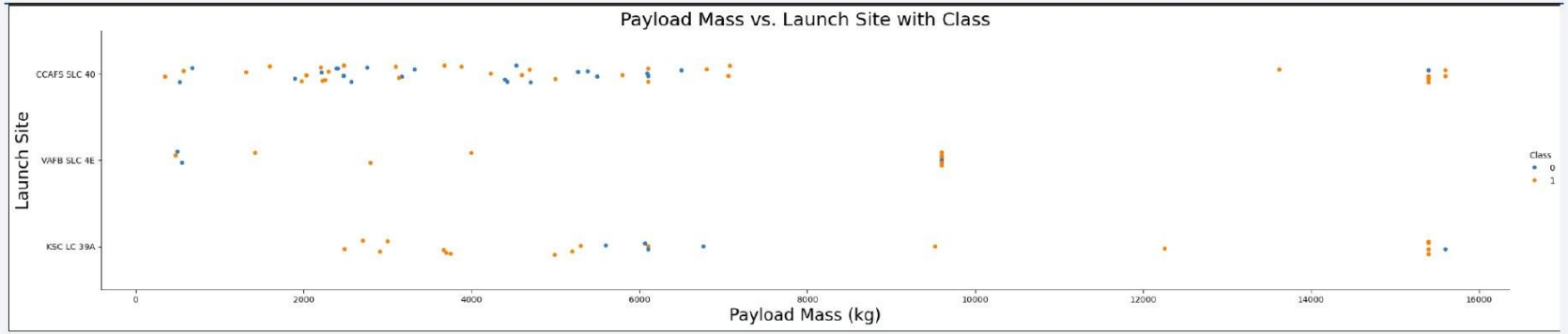
CCAFS SLC 40: This launch site shows a high concentration of early flights, particularly those with a 'Class' of 0 (unsuccessful landing). As the flight number increases, there's a clear trend towards more successful landings (Class 1) at this site.

VAFB SLC 4E: This site has fewer launches overall. For early flight numbers, success rates appear mixed, but with increasing flight numbers, there's a tendency towards successful landings.

KSC LC 39A: This site seems to have been introduced later in the sequence of flight numbers. It generally shows a very high success rate from its inception, with most launches resulting in successful landings.

In general, across all sites, there's a visible trend: as the flight number increases, the probability of a successful landing (Class 1) also increases, indicating improved technology, experience, and reliability over time. Different launch sites also appear to have different success rates, possibly due to factors like their operational history and the type of missions they typically host.

Payload vs. Launch Site



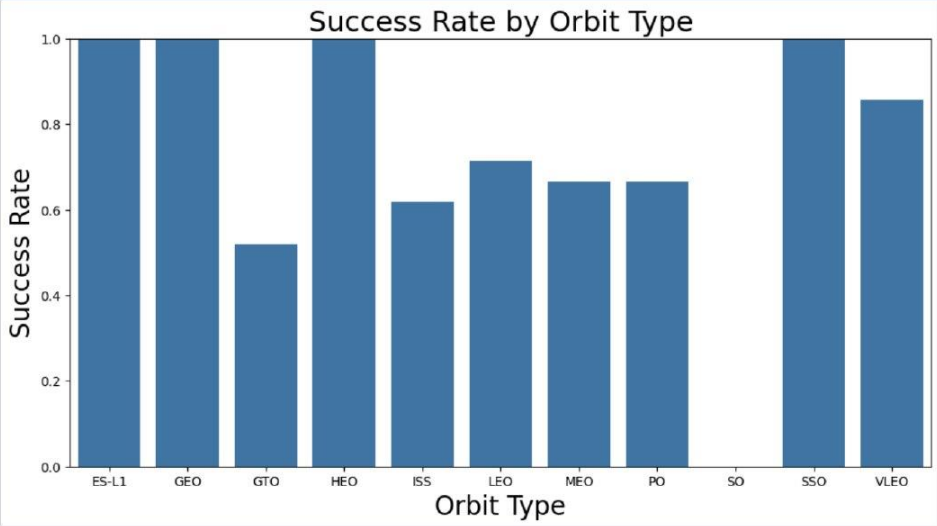
VAFB SLC 4E (Vandenberg Space Force Base, Space Launch Complex 4E): this launch site primarily handles payloads with masses generally below 10,000 kg. There are indeed no rockets launched for heavy payload mass (greater than 10,000 kg) from this site. This might suggest a focus on specific types of missions, possibly polar orbits, which often involve lighter satellites.

CCAFS SLC 40 (Cape Canaveral Space Force Station, Space Launch Complex 40): This site handles a wide range of payload masses, including many heavy payloads. It shows a good mix of both successful (Class 1) and unsuccessful (Class 0) landings across various payload masses, but with a general trend towards higher success rates for heavier payloads in later flights.

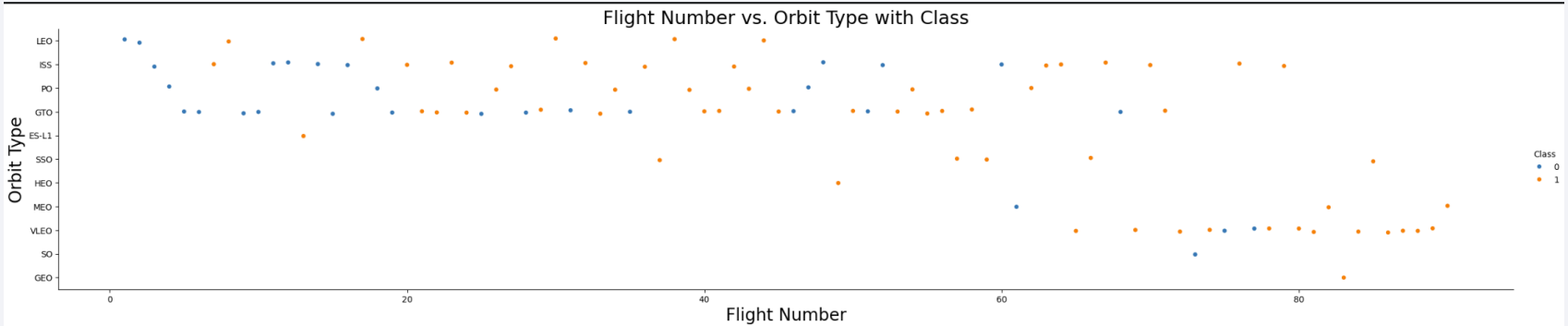
KSC LC 39A (Kennedy Space Center, Launch Complex 39A): This site also accommodates a broad spectrum of payload masses, including very heavy ones. Notably, KSC LC 39A demonstrates a very high success rate for almost all payload masses, indicating its capability and reliability for diverse and often heavier missions.

Overall, the plot indicates that launch sites have different operational envelopes in terms of payload mass. While some sites like CCAFS and KSC handle a full range, VAFB-SLC appears to specialize in lighter payloads. The success rate (Class) also varies by launch site and payload mass, with heavier payloads generally showing better success in later flights at the more active sites.

Success Rate vs. Orbit Type

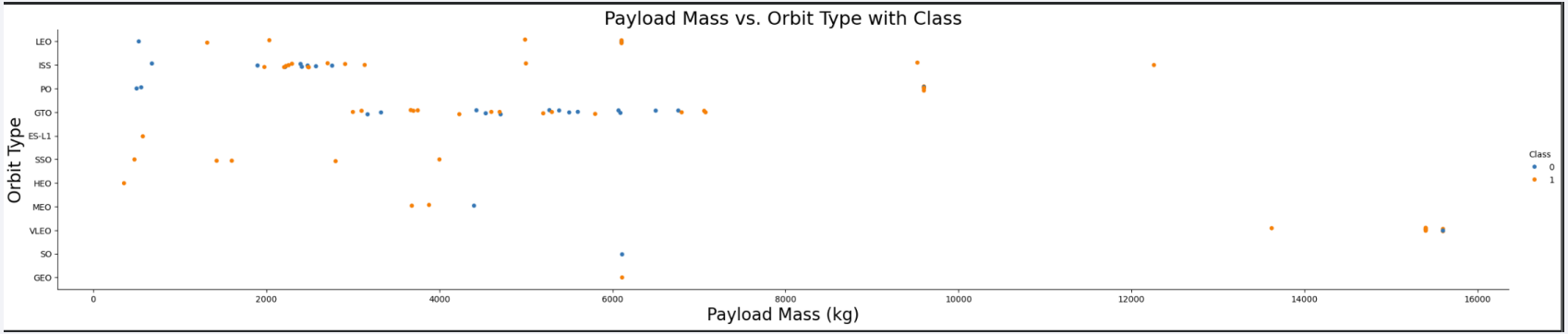


Flight Number vs. Orbit Type



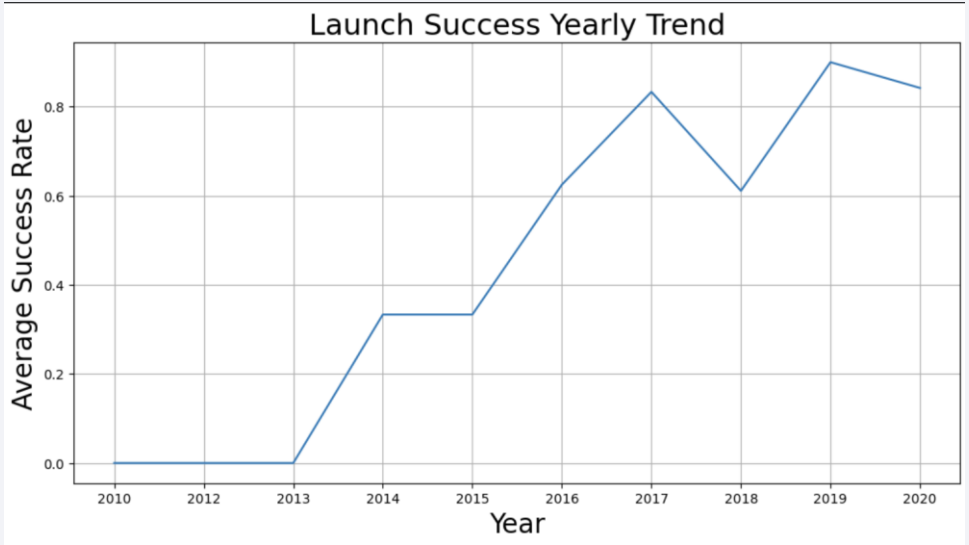
- **LEO:** success improves clearly with flight number; later LEO flights almost all landed
- **GTO:** successes and failures appear throughout, with no clear link to flight number
- **ISS:** mixed outcomes; 62% of ISS missions landed (13 of 21)
- **Other orbits** have few launches each, so their rates are not reliable on their own

Payload vs. Orbit Type



- **Polar, LEO and ISS:** heavy payloads land successfully more often
- **GTO:** successes and failures are mixed across payload masses, so payload alone does not predict landing
- **Overall:** the effect of payload depends on the orbit, which is why both features are kept for modelling

Launch Success Yearly Trend



- **2010–2013:** no first stage landed (early experimental attempts)
- **2014–2017:** success rose from 33% to 83%
- **2018:** a dip to 61%
- **2019–2020:** 90% and 84%
  - Landing reliability improved strongly with experience, but it had not reached 100% by 2020

All Launch Site Names

```
%%sql
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
SELECT SUM(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(Payload_Mass__kg_)
45596

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(Payload_Mass__kg_)
2534.6666666666665

First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
%%sql
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND Payload_Mass__kg_ > 4000 AND Payload_Mass__kg_ < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
%%sql
SELECT Mission_Outcome, COUNT(Mission_Outcome) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%%sql
SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Payload_Mass__kg_ = (SELECT MAX(Payload_Mass__kg_) FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%%sql
SELECT
  substr(Date, 6, 2) AS Month,
  Booster_Version,
  Launch_Site
FROM SPACEXTABLE
WHERE
  substr(Date, 0, 5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT Landing_Outcome, COUNT(Landing_Outcome) AS Total_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Total_Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

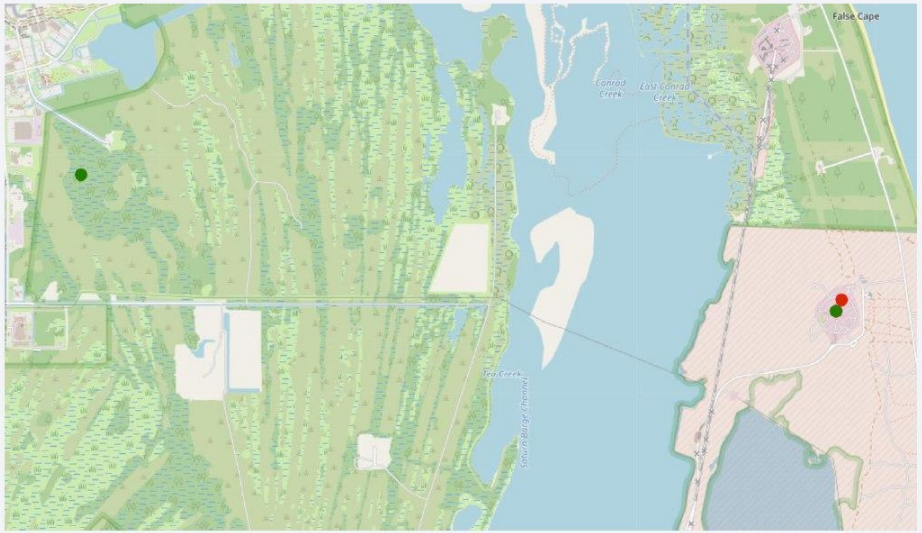
Landing_Outcome	Total_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1



Section 3

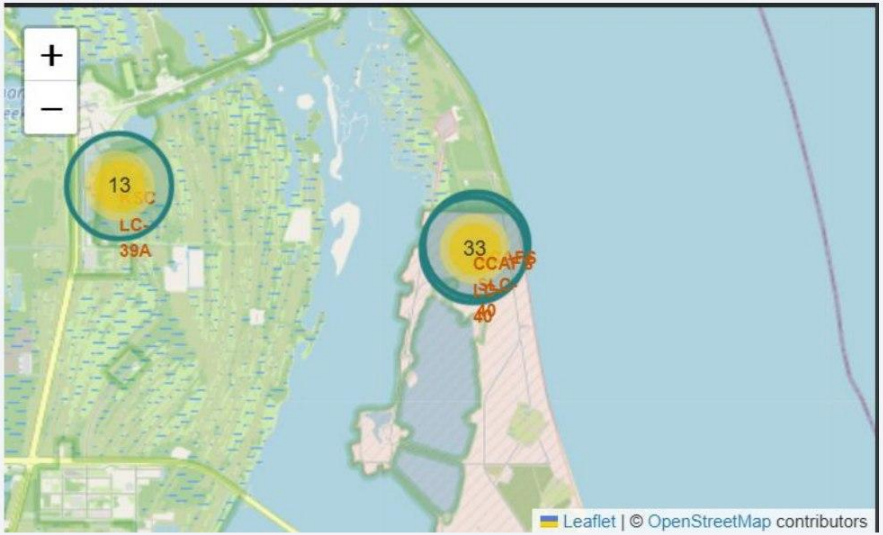
Launch Sites Proximities Analysis

Folium Map Showing all Launch Sites location marker

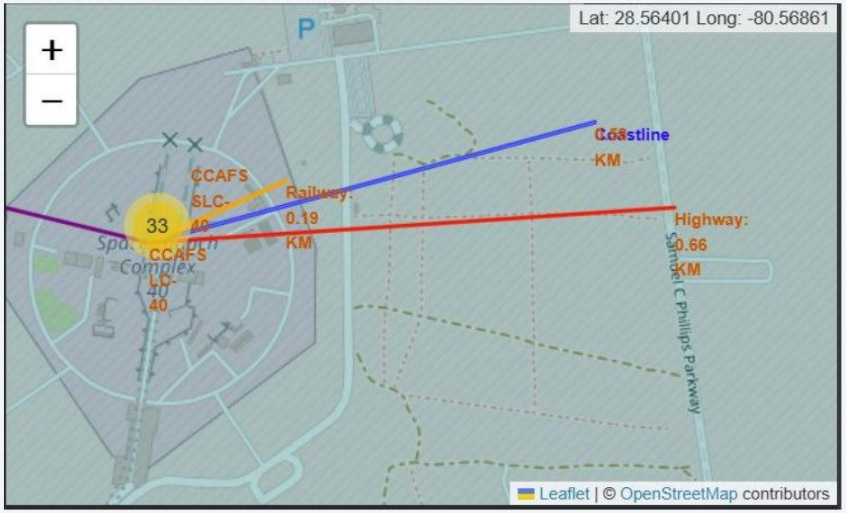


	Launch Site	Lat	Long	class
46	KSC LC-39A	28.573255	-80.646895	1
47	KSC LC-39A	28.573255	-80.646895	1
48	KSC LC-39A	28.573255	-80.646895	1
49	CCAFS SLC-40	28.563197	-80.576820	1
50	CCAFS SLC-40	28.563197	-80.576820	1
51	CCAFS SLC-40	28.563197	-80.576820	0
52	CCAFS SLC-40	28.563197	-80.576820	0
53	CCAFS SLC-40	28.563197	-80.576820	0
54	CCAFS SLC-40	28.563197	-80.576820	1
55	CCAFS SLC-40	28.563197	-80.576820	0

Folium Map – Coloured Label outcomes of sites



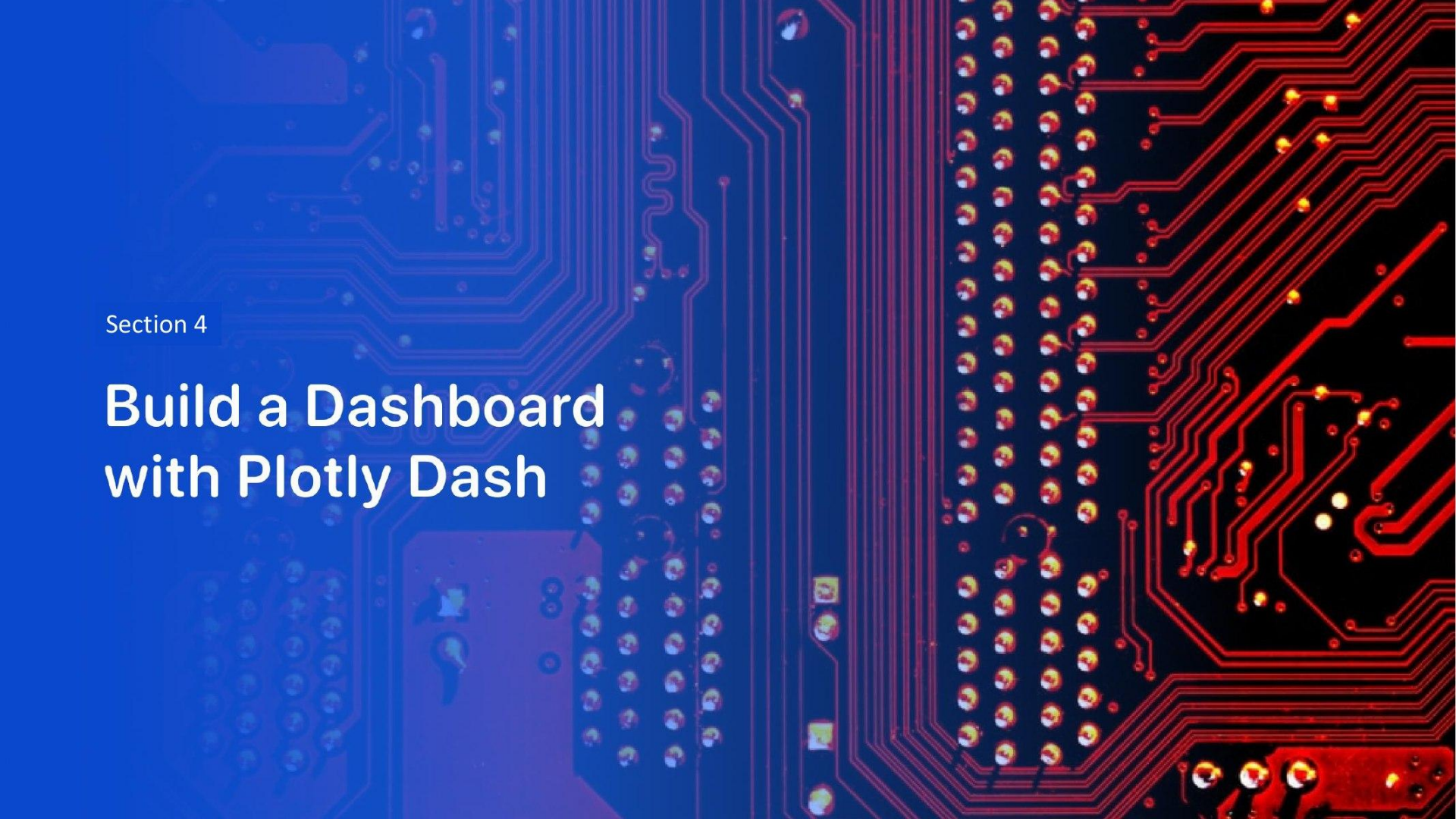
Folium Map showing marker with distance to a closest city, railway, highway,Coast line



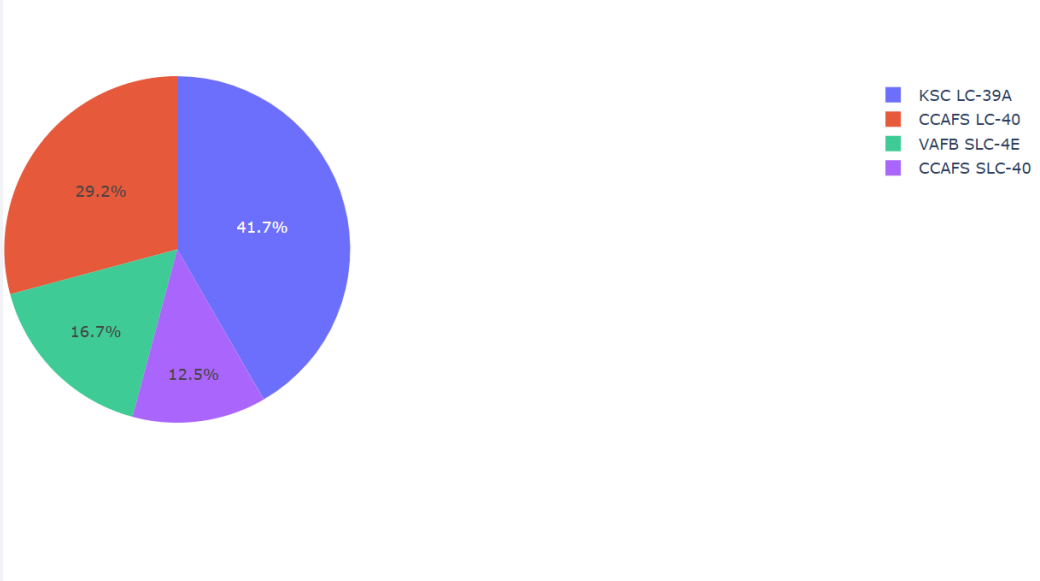


Findings:

Distance from CCAFS LC-40 to City: 23.17 km
Distance from CCAFS LC-40 to Railway: 0.19 km
Distance from CCAFS LC-40 to Highway: 0.66 km
Distance from CCAFS LC-40 to coastline: 0.58 km

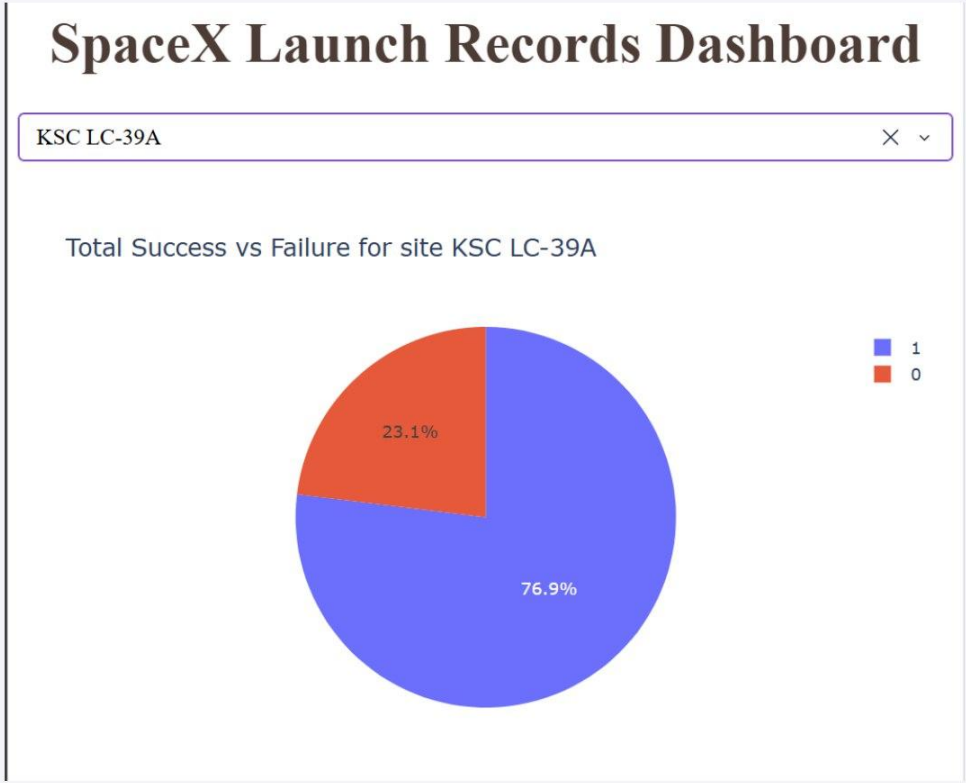

Section 4
Build a Dashboard with Plotly Dash

Total Success Launches by Site Record Dashboard



- **KSC LC-39A** has the largest share of successful launches (41.7%) and the highest success rate (76.9%)
- **CCAFS LC-40** hosted the most launches in this dataset (26) but had the lowest success rate (26.9%), as it carried most early flights

SpaceX Launch Records Dashboard



- **Operational Insight**: The site's infrastructure supports a high success rate, but the failure percentage highlights the challenges of complex missions often attempted from LC-39A. LC-39A is **reliable with a 3:1 success-to-failure ratio**, but the 23.1% failure share reminds us that it's also the site of some of SpaceX's most ambitious and risk-heavy missions.

Payload and Success for All sites Dashboard

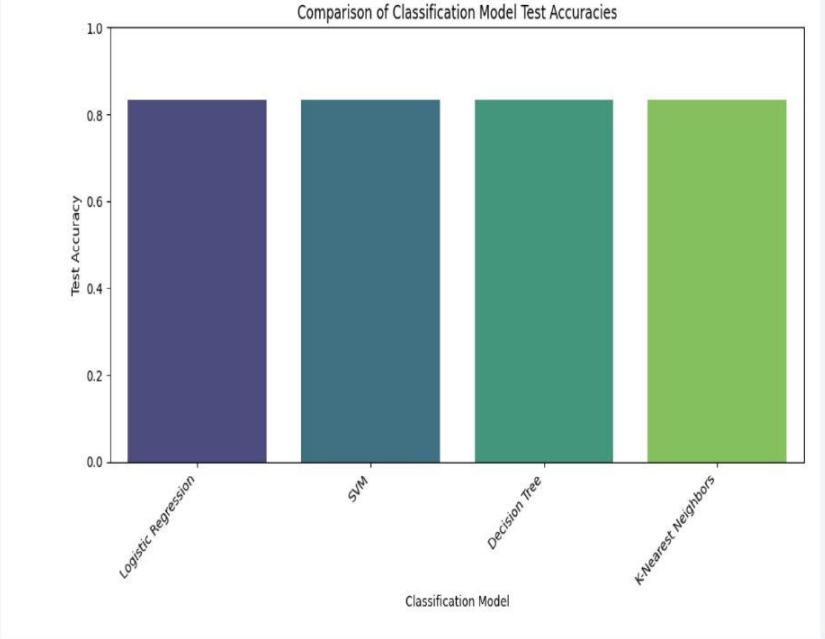


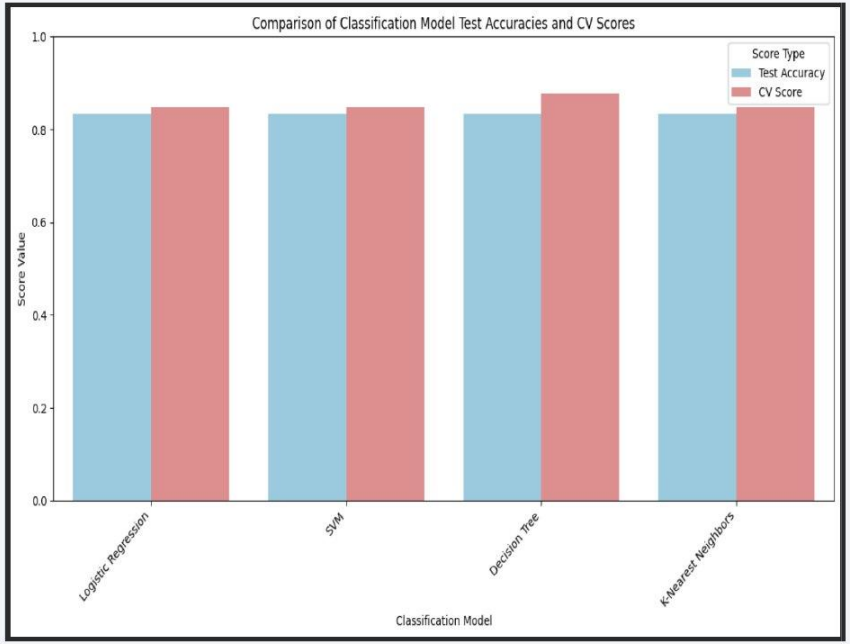
Section 5

Predictive Analysis (Classification)

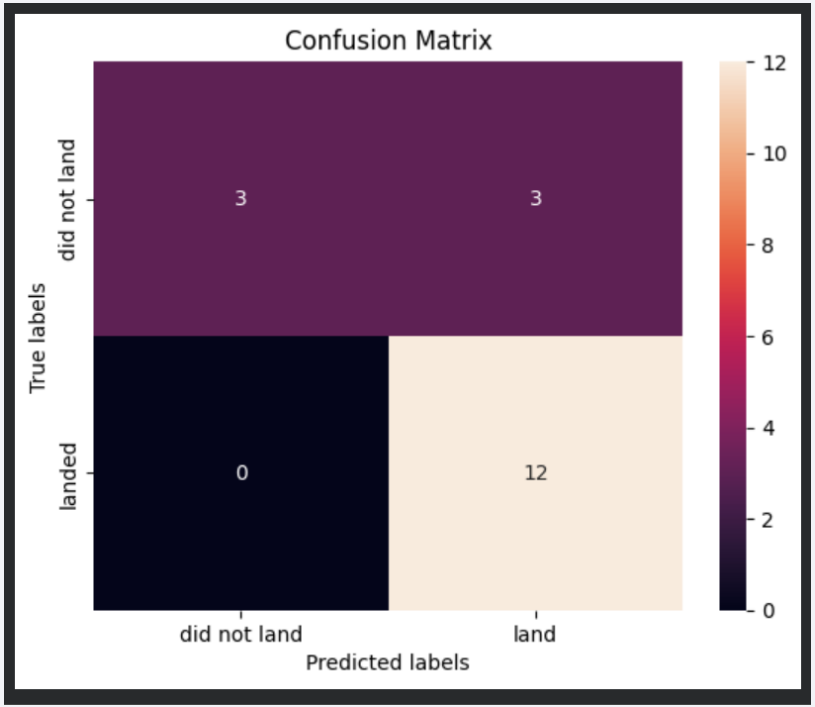


Classification Accuracy





Confusion Matrix



- All four models produce this same confusion matrix on the 18 test launches; the Decision Tree is shown because it has the best CV accuracy
- **True positives = 12:** every booster that landed was predicted correctly
- **True negatives = 3:** 3 of 6 failed landings were predicted correctly
- **False positives = 3:** the model predicted a landing that did not happen
- **False negatives = 0:** recall for landings is 100%
- Accuracy = 15 / 18 = 0.833

Conclusions

- **Experience drives success:** the first 25 flights landed 36% of the time; later flights landed 78.5% of the time
- **Orbit matters:** VLEO (86%) and LEO (71%) land most often; GTO is lowest (52%). ES-L1, GEO, HEO and SSO show 100% but on 1–5 launches each
- **Payload matters by orbit:** heavy payloads land more often for Polar, LEO and ISS missions; GTO shows no clear pattern
- **Launch sites:** VAFB SLC-4E never launched payloads above 10,000 kg; KSC LC-39A has the highest success rate
- **Site location:** CCAFS LC-40 is close to a railway (0.19 km), highway (0.66 km) and coastline (0.58 km) and 23 km from the nearest city

Conclusions

- **Model:** all four classifiers reach 83.3% test accuracy; the Decision Tree has the best CV accuracy (88.8%). The only errors are false positives
- **SQL facts:** first ground-pad landing on 2015-12-22; two drone-ship landing failures in 2015 (January and April); the maximum payload was 15,600 kg
- **Limitations:** 90 launches and an 18-launch test set are too few to separate the models; a different random split could change the ranking
- **Next steps:** repeated cross-validation, more recent launch data, and features such as booster reuse count and landing type

Appendix

- **Repository:** github.com/Sumant-crty/WinningSpaceRaceWithDataScience
- **Notebooks:** notebooks/
- **Dashboard app:** dashboard/spacex_dash_app.py
- **Data sources:** SpaceX REST API v4; Wikipedia "List of Falcon 9 and Falcon Heavy launches" (fixed snapshot); IBM Skills Network course datasets
- **Tools:** Python, pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Folium, Plotly Dash, BeautifulSoup, SQLite

Thank you!

